
EEE5776. Applied Machine Learning



Module 2. Supervised Learning and Experimental Design

Classification with Logistic Regression
The Objective Function
Gradient Descent
Implementation

Introduction

When we introduced supervised learning, we discussed two main tasks:

- regression (predicting values) and
- classification (predicting classes).

Classifiers can be categorized as **discriminative** or **generative** classifiers. And sometimes a **hybrid** version probabilistic discriminative classifier, such as the **logistic regression** algorithm.

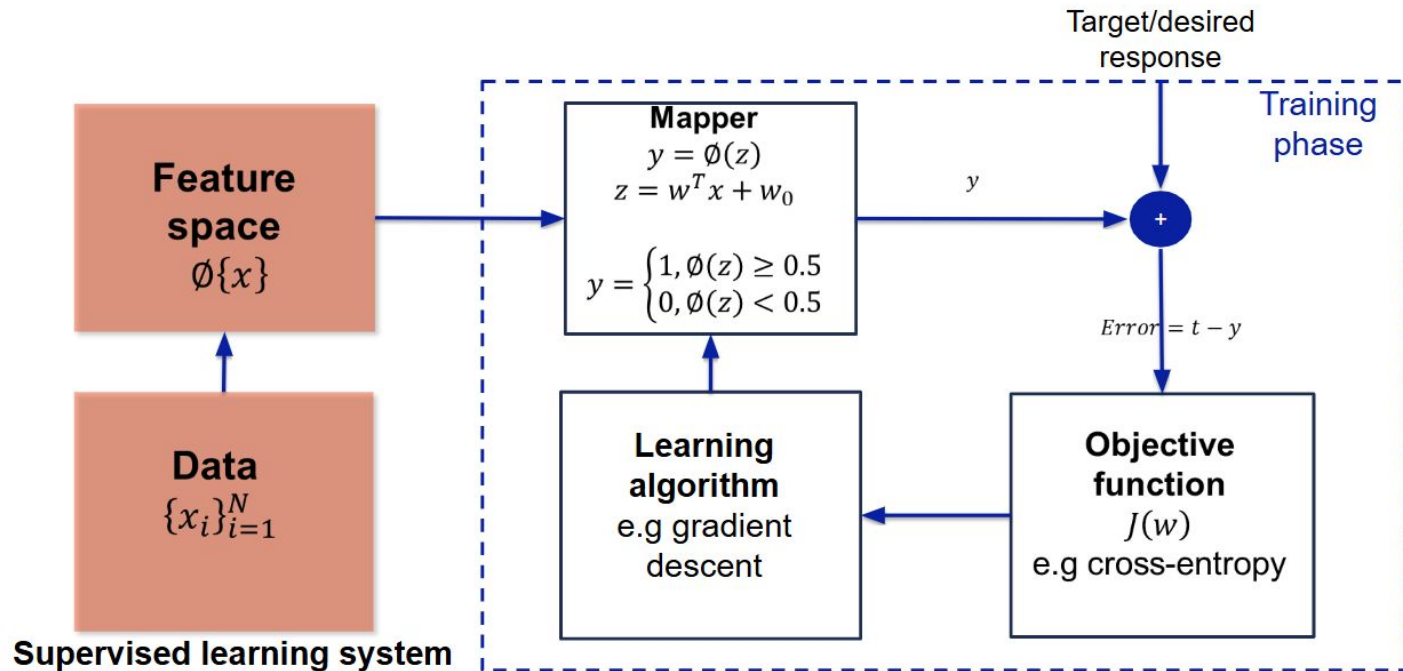
Introduction

- A **discriminative** approach for classification is one in which we partition the feature space into regions for each class.
 - Then, when we have a test point, we evaluate in which region it landed on and classify it accordingly.
- A **generative** approach for classification is one in which we estimate the parameters for distributions that generate the data for each class using Bayesian principles.
 - When we have a test point, we can compute the posterior probability of that point belonging to each class and assign the point to the class with the highest posterior probability.

Logistic Regression

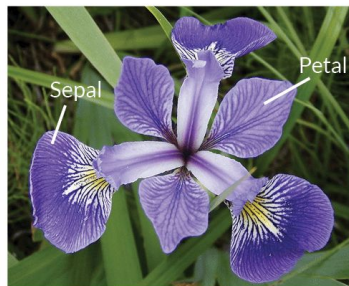
- Note that, despite of its name, logistic regression is a model for classification, not regression.
- The logistic discriminant is a linear model for binary classification that can be extended to multi-class classification using the one-vs-all approach.
- Logistic regression is a probabilistic discriminative classifier.

Supervised Learning Process - Logistic Regression Example



Example

- Iris plants dataset
- Number of Instances: 150 (50 in each of three classes)
- Number of Attributes: 4 numeric,
- Attribute Information: - sepal length in cm - sepal width in cm - petal length in cm - petal width in cm - class: - Iris-Setosa - Iris-Versicolour - Iris-Virginica



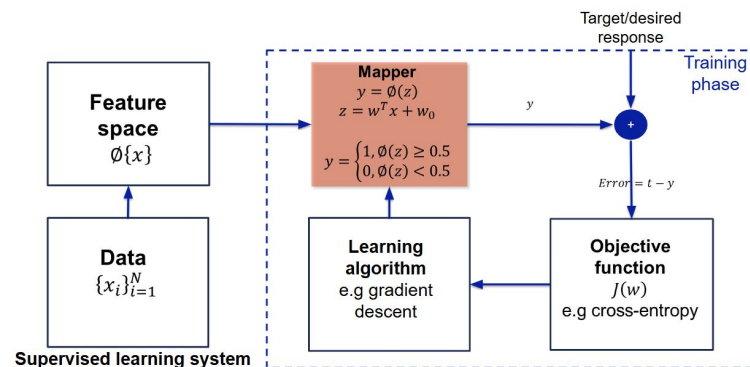
Iris Versicolor



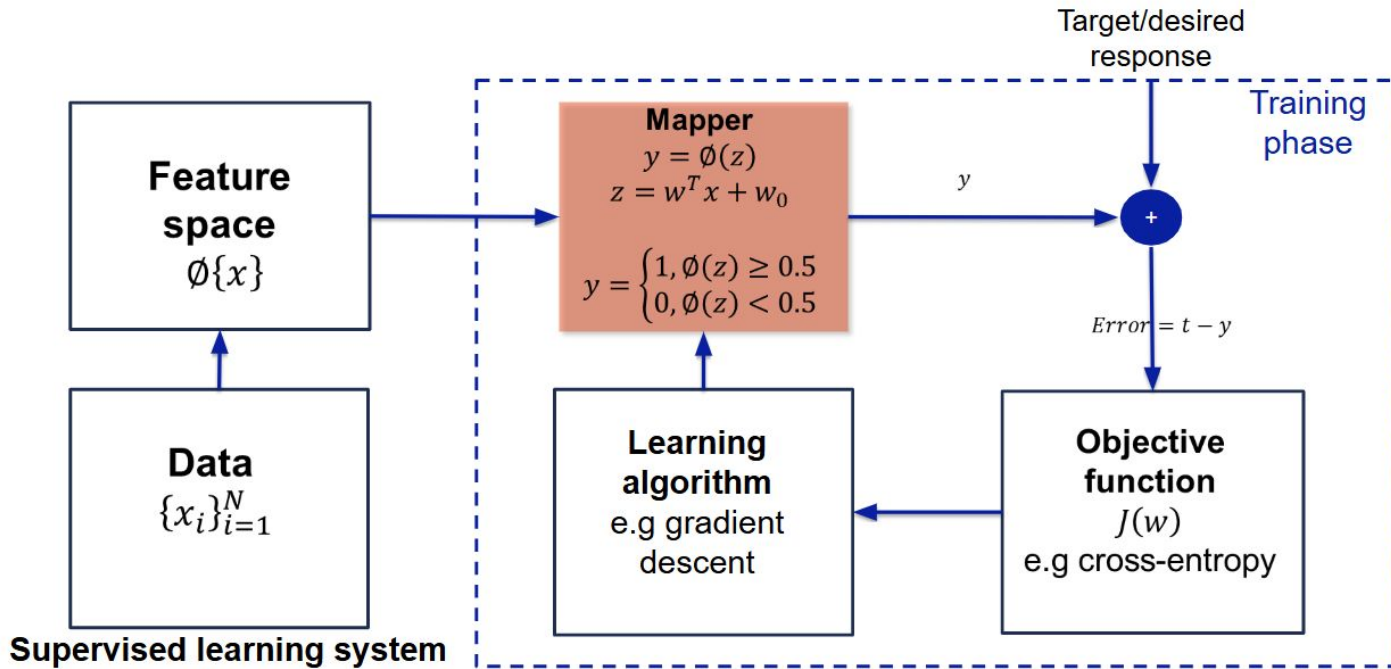
Iris Setosa



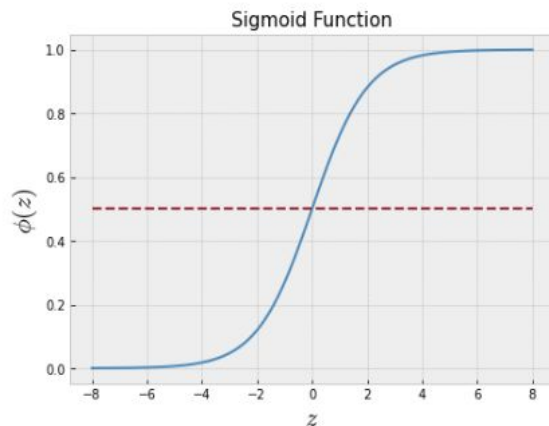
Iris Virginica



Supervised Learning Process - Logistic Regression Example

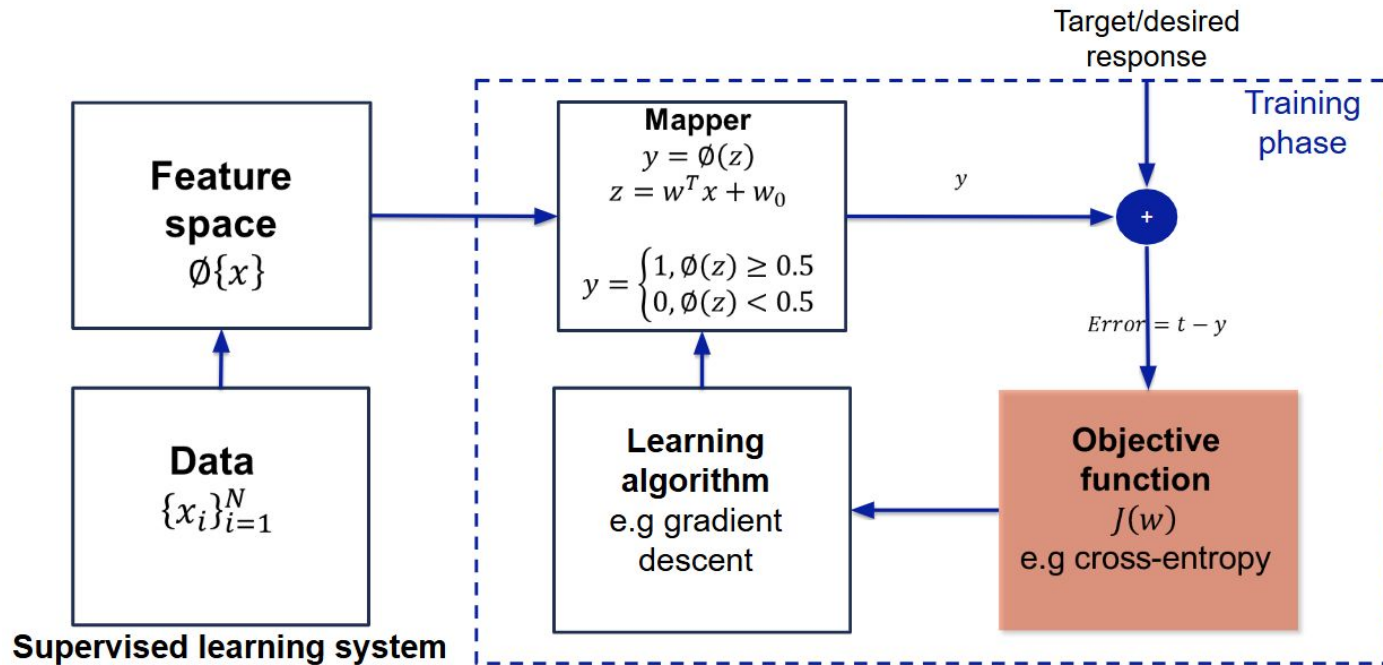


The Mapper Function

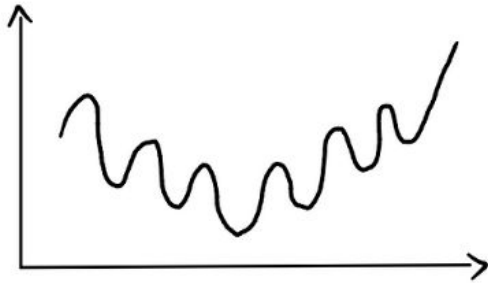


$$\hat{p} = y = \begin{cases} 1, & \phi(z) \geq 0.5 \\ 0, & \phi(z) < 0.5 \end{cases}$$

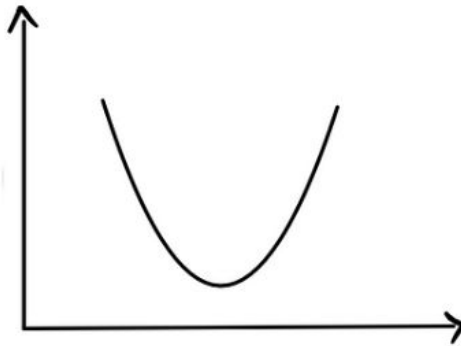
Supervised Learning Process - Logistic Regression Example



The Objective Function



If we use the MSE – non-convex
Difficult to optimize!

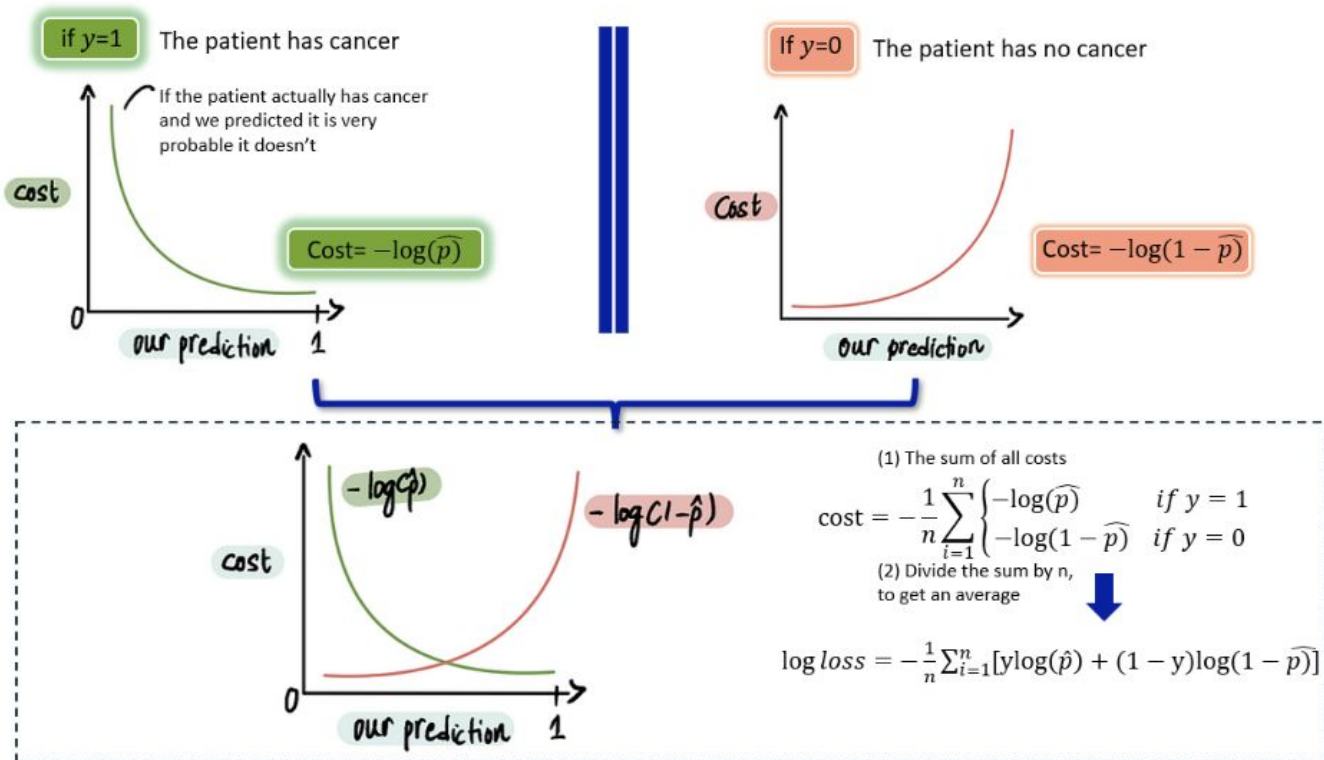


How to achieve a pretty, convex
curve such as this one?

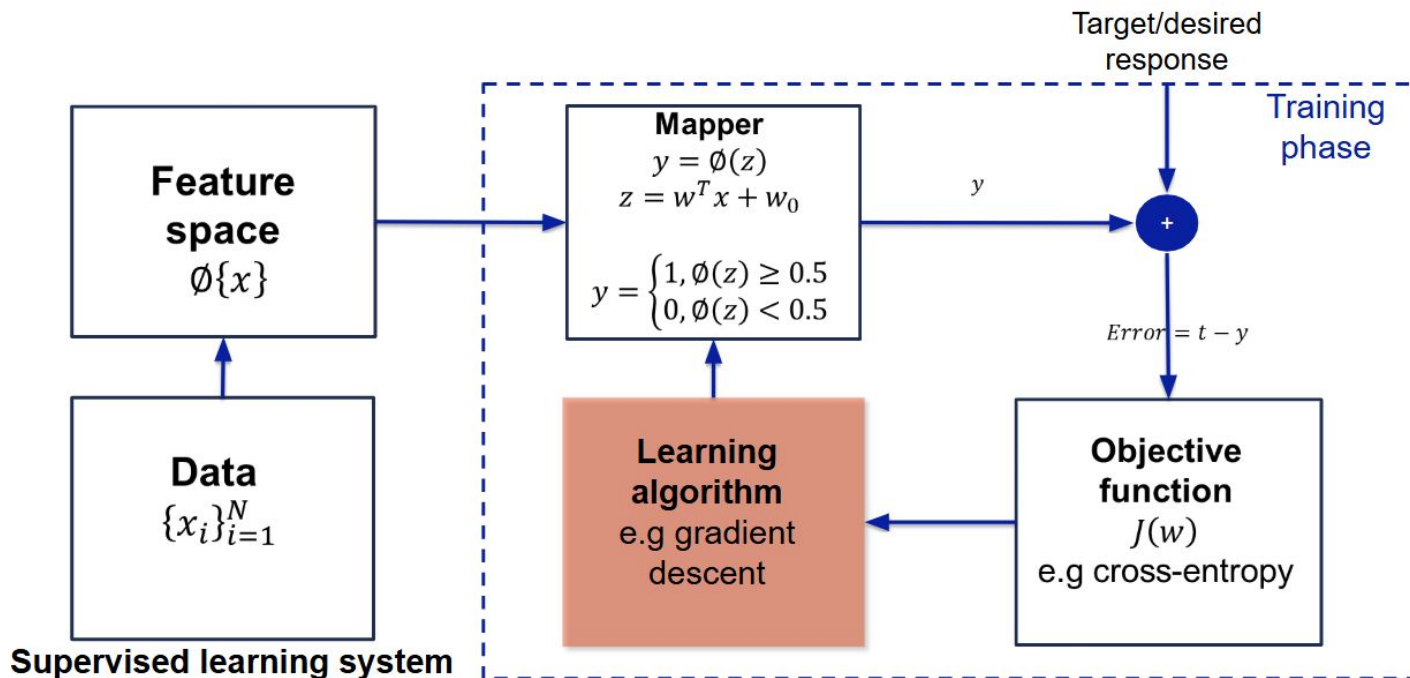


Use the Logistic (Log) Loss
Cost Function

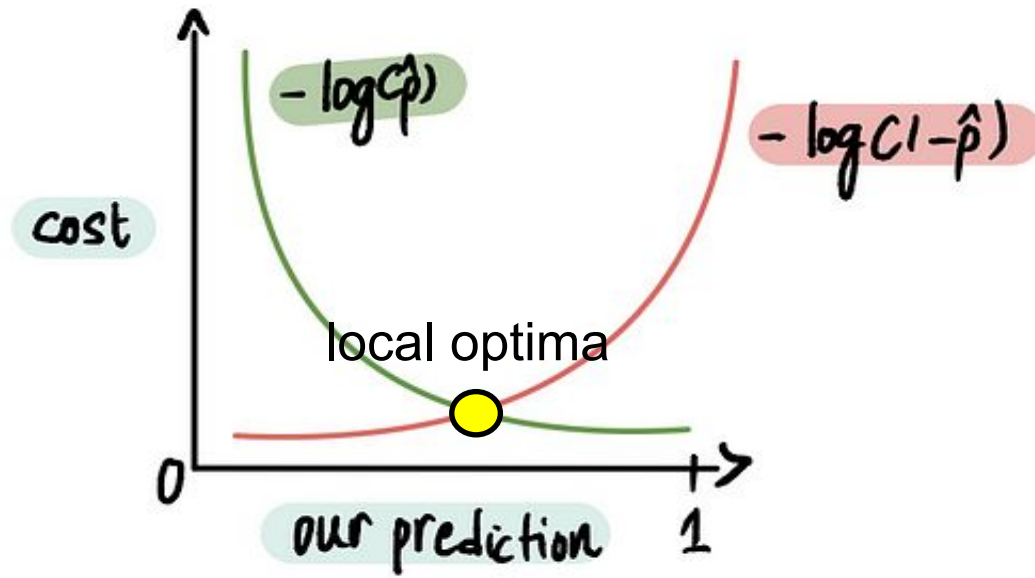
The Objective Function



Supervised Learning Process - Logistic Regression Example



Supervised Learning Process - Logistic Regression Example



Weight Initialization

- **Initialization:** it is best to initialize w with random values close to 0; generally they are drawn uniformly from the interval $[-0.01, 0.01]$.
- The reason for this is that if the initial w are large in magnitude, the weighted sum may also be large and may saturate the sigmoid.
- If the initial weights are close to 0, the sum will stay in the middle region where the derivative is nonzero and an update can take place.
- If the weighted sum is large in magnitude (smaller than -5 or larger than $+5$), the derivative of the sigmoid will be almost 0 and weights will not be updated.
 - When we stack up a lot of these perceptrons in layers and add a few layers, this "close to 0" gradient will lead to a phenomenon in neural networks known as the **vanishing gradient**.



Training a Binary Classifier

Jupyter Notebook

- Follow the code included in the Jupyter Notebook for today's class. It is on the GitHub repository under Materials → Notebooks → Session 8. Classification using Logistic Regression (02-11).ipynb